

Sanjana Singamsetty

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SUMMARY

Software Engineer at PayPal with 1+ year of experience designing and building scalable, production-grade web applications using **React**, **TypeScript**, **JavaScript**, and modern frontend frameworks. Published AI researcher with four peer-reviewed IEEE and PeerJ publications spanning Machine Learning, Natural Language Processing, Explainable AI, Healthcare AI, and Time Series Forecasting. College Third Rank Holder and Gold Medalist in Artificial Intelligence & Machine Learning. Skilled in full-stack development, REST API integration, AI/ML model deployment, and Agentic AI workflows. Passionate about building AI-powered products, scalable software systems, and intelligent full-stack applications.

WORK EXPERIENCE

Software Engineer — PayPal

Jul 2025 – Present

- Delivered scalable, high-priority frontend applications under tight deadlines using **React**, **TypeScript**, **JavaScript**, **HTML5**, and **CSS3**, following Agile development practices.
- Served as on-call engineer, performing production debugging, root-cause analysis, and incident resolution to ensure reliability of customer-facing systems.
- Built **Agentic AI** workflows and automation using **MCP (Model Context Protocol)**, improving developer productivity and reducing manual effort.
- Engineered reusable, accessible UI components aligned with **WCAG** standards and responsive design principles, improving maintainability across the codebase.
- Collaborated cross-functionally to integrate **REST APIs**, automate workflows, and ship production-ready features using **Git**, **CI/CD**, and Agile methodologies.

Freelance Frontend Developer — Doon International School, Hyderabad

2025

- Designed and developed the official school website with a focus on responsiveness, accessibility, and user experience (UX).
- Built reusable frontend components and optimized site performance across desktop and mobile platforms.
- Partnered directly with stakeholders to gather requirements, implement design improvements, and ship production-ready solutions.
- Managed end-to-end deployment, maintenance, and post-launch feature enhancements.

AWS Intern — APSSDC

May 2023 – Aug 2023

- Deployed and managed applications using **AWS** cloud services, gaining hands-on experience with cloud infrastructure and CI/CD deployment workflows.

Flutter Developer Intern — Bragtech

May 2023 – Aug 2023

- Developed cross-platform mobile applications using **Flutter** and **Firebase**, implementing responsive, production-ready user interfaces.

Android Developer Intern — Aerial Topographic Services Pvt. Ltd.

Mar 2023 – Apr 2023

- Built native Android applications with **Firebase** integration, implementing secure authentication and CRUD functionality.

ACHIEVEMENTS

- **Gold Medalist – Artificial Intelligence & Machine Learning**
- **College Third Rank Holder** with CGPA: 9.48/10.

- **Gold Medal – Research Day** for *Enhancing Machine Learning-based Forecasting of Chronic Renal Disease with Explainable AI*.
- **First Prize – Mahindra University Research Symposium** for patented innovation recognized for originality and impact.

PUBLICATIONS (4 PEER-REVIEWED)

Comparative Analysis of Time Series Forecasting Models: Evaluating Hybrid Approaches and FB Prophet for Optimal Performance 2025
S. Singamsetty, M. Chimata, J. L. Manohar, M. S. Nandan, M. K. Enduri — CONIT, IEEE.

Enhancing Machine Learning-based Forecasting of Chronic Renal Disease with Explainable AI 2024
S. Singamsetty, S. Ghanta, S. Biswas, A. Pradhan — PeerJ Computer Science, 10:e2291.

Employing TF-IDF and Word2Vec Embeddings to Identify Multi-Class Toxicity Through Machine and Deep Learning Approaches 2024
S. Singamsetty, H. Somayajula, A. L. Immadisetty, K. S. Konkimalla, S. Tokala, M. K. Enduri — OCIT, IEEE.

Predicting Machine Learning for Early Diabetes Mellitus Prediction 2024
S. Singamsetty, M. S. Nandan, S. S. C. Polu, J. L. Manohar, S. Tokala, M. K. Enduri — BTS-I2C, IEEE.

SELECTED PROJECTS

SmartVisualizer – AI-Powered GitHub Repository Explorer[GitHub](#) [Live Demo](#)

Built a full-stack AI tool that analyzes any public GitHub repository in seconds and lets users chat with an AI that has read every file. Features an interactive, color-coded file-tree **Structure Explorer**, commit-activity bubble charts (**Commit Analysis**), **Branch Overview** with AI-generated branching strategy summaries, language/tech-stack **Code Insights**, an **AI Chat** agent with Deep Search for precise file-level answers, and an AI-summarized **README Viewer**. Built with **React**, **TypeScript**, AI/LLM integration, and the GitHub API; deployed on **Vercel**.

React Learning – Full-Stack ML Web App[GitHub](#) [Live Demo](#)

Built a full-stack ML-enabled web application using **React**, **Node.js**, and **Express**, integrating secure user data handling, REST APIs, and an interactive UI; used **BullMQ** and **Redis** for background job queuing and asynchronous task processing; deployed and continuously updated on **Vercel**.

SKILLS

Programming Languages	Python • C++ • JavaScript • TypeScript
Frontend	React.js • Next.js • HTML5 • CSS3 • Tailwind CSS • Redux
Backend	Node.js • Express.js • REST APIs • BullMQ • Redis
Databases	MongoDB • Firebase
AI & Machine Learning	Machine Learning • NLP • Explainable AI • GenAI • LLMs • Agentic AI • Transformers • Hugging Face • TensorFlow • Scikit-learn • Pandas • NumPy • Streamlit
Cloud & DevOps	AWS • Git • GitHub • Docker • CI/CD • Vercel • Postman
Core CS	Data Structures & Algorithms • OOP • Operating Systems • Computer Networks • System Design
Interests: Artificial Intelligence • Full Stack Development • System Design	

EDUCATION

2021 – 2025	Bachelor of Technology (Computer Science Engineering) , SRM AP University (CGPA: 9.48/10) College Third Rank Holder — Gold Medalist – Artificial Intelligence & Machine Learning	
2019 – 2021	Class XII, State Board – Chaitanya College	(954/1000)
2019	Class X, CBSE – Atkinson Senior Secondary School	(384/500)